PGPM 2026–27 | Business Analytics Project

Predicting and Explaining Employee Attrition

A Consulting Analysis of Retention Risk and Career - Growth Dynamics

Field	Detail
Course	Business Analytics
Data Source	HR Analytics Employee Attrition Dataset
Group	A4
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1. Executive Summary

Employee attrition is a direct, measurable cost - every voluntary exit triggers recruitment spend, onboarding time and a loss of institutional knowledge. Using the IBM HR Analytics dataset (1,470 employees across Sales, Research & Development and Human Resources), this project set out to identify who is leaving, why and where HR should concentrate its retention effort.

Overall attrition stands at 16.1%, but that average hides sharp concentration. Sales Representatives leave at nearly 40% more than double the next riskiest role and Sales and HR run attrition roughly 1.4-1.5 times higher than R&D. Employees who leave tend to have shorter tenure, lower job level and income, no stock options and are disproportionately on frequent travel or working overtime.

We built a logistic regression model to interpret these drivers. It shows around 85.3% test accuracy, but that number is misleading on its own: simply predicting "nobody leaves" already scores 83.9%, because only 1 in 6 employees actually attrits. The more honest metric is recall on actual leavers, and every model we tested caught less than half of them, logistic regression performed best at 39%. This tells us the available data separates roughly 5 in 6 employees who stay reasonably well, but the leavers are not cleanly distinguishable on these variables alone.

Separately, to examine the career-progression concern flagged in HR's Career Stagnation Index, we used composite Career Growth Score from salary hikes, tenure and promotion timing and regressed it against those same drivers. The model explains just over half the variance ($R^2 \approx 0.51 - 0.56$), with time since last promotion emerging as by far the strongest negative pull-on career growth, which is a clearer, more actionable signal than the attrition model alone provides.

Together, the findings point HR away from a single blanket retention policy and toward two specific levers: role-targeted intervention in Sales (and to a lesser extent HR) and closer management of the promotion cycle as a leading indicator of disengagement well before an employee formally resigns.

Also, we did not use clustering approach as it is not suitable for addressing our problem statement.

2. Problem Statement & Research Questions

Attrition is treated here as a management decision problem rather than a pure classification exercise: which factors should HR act on, where should retention effort be concentrated and does a model built on this data actually help target that effort? The interim proposal for this project set out five linked business questions; we consolidate them into four testable research questions for the final analysis.

Research Question	Data Used	Method
RQ1: Which employee attributes are most associated with attrition?	All 31 usable employee-level attributes (demographic, compensation, tenure, satisfaction)	Correlation analysis + logistic regression on standardized features
RQ2: Can a model reliably flag at-risk employees?	70/30 stratified train-test split	Logistic regression

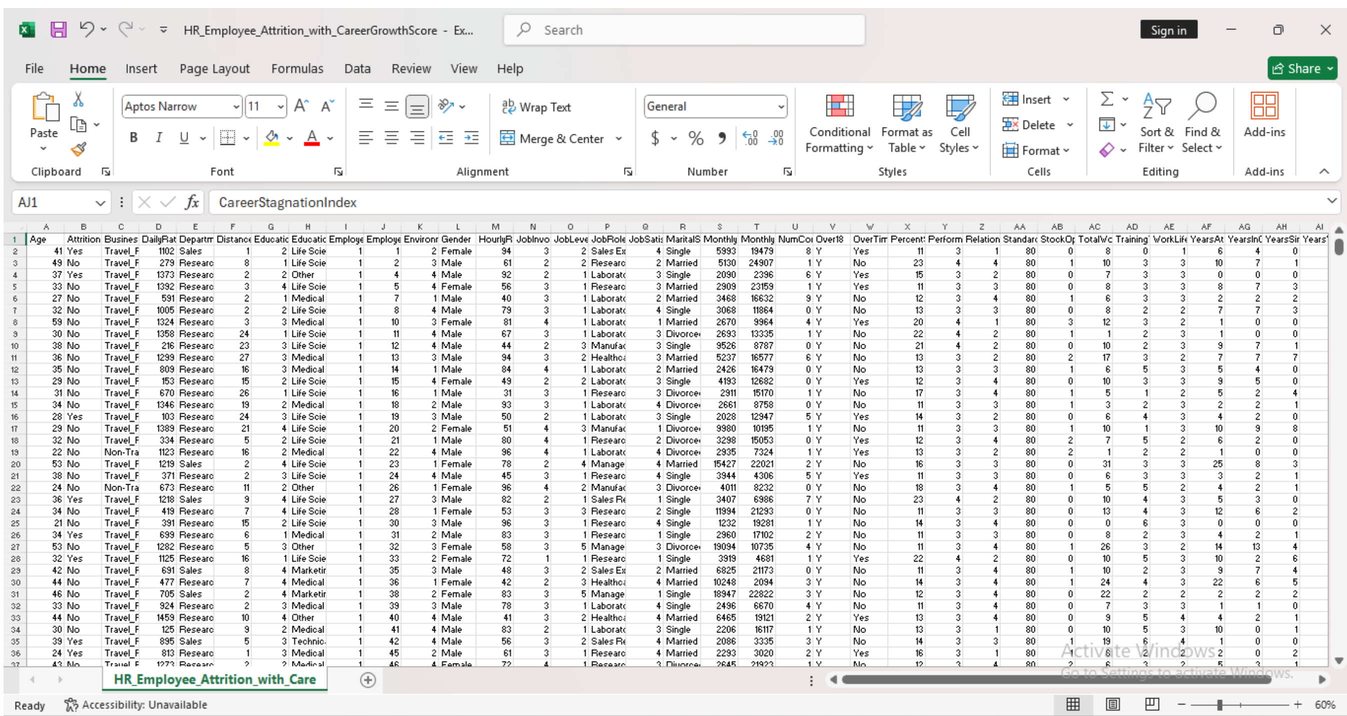
Research Question	Data Used	Method
RQ3: Which department / job role carries the most risk?	Department and Job Role vs. Attrition	Group-wise attrition-rate computation
RQ4: Are there measurable career-progression dynamics HR could track as an early-warning signal?	PercentSalaryHike, TotalWorkingYears, YearsAtCompany, YearsSinceLastPromotion, CareerStagnationIndex	Linear regression

RQ1–RQ3 map directly onto the attrition classification problem defined in the interim report. RQ4 extends the analysis: rather than only asking who leaves, we test whether the same underlying drivers (pay growth, tenure, promotion delay) can be summarized into a single career-growth metric management could monitor continuously rather than waiting for a resignation event.

3. Data & Methodology

3.1 Data Source and Structure

The analysis uses the IBM HR Analytics Employee Attrition dataset (Kaggle), extended with a pre-built Career Stagnation Index. Each of the 1,470 rows represents one employee at one point in time with 37 columns spanning demographics, job and organizational structure, compensation, performance and satisfaction survey results and tenure history. The dataset is fully populated and there are no missing values to impute.



3.2 Data Preparation

Three columns were constant across all 1,470 employees (EmployeeCount, Over18, StandardHours) and one (EmployeeNumber) was a pure identifier, all four were dropped as they carry no analytical signal. The target variable is Attrition which was label-encoded (Yes = 1, No = 0). Remaining categorical variables (Department, JobRole, MaritalStatus, BusinessTravel, OverTime, EducationField, Gender) were one-hot encoded for modeling, and duplicate engineered columns were removed. No original column was altered, all feature engineering was appended as new columns.

3.3 Feature Engineering: Career Growth Score

For RQ4, we used composite Career Growth Score as a transparent, weighted combination of four progression signals - PercentSalaryHike, TotalWorkingYears, YearsAtCompany and YearsSinceLastPromotion - together with the provided CareerStagnationIndex.

3.4 Analytical Techniques

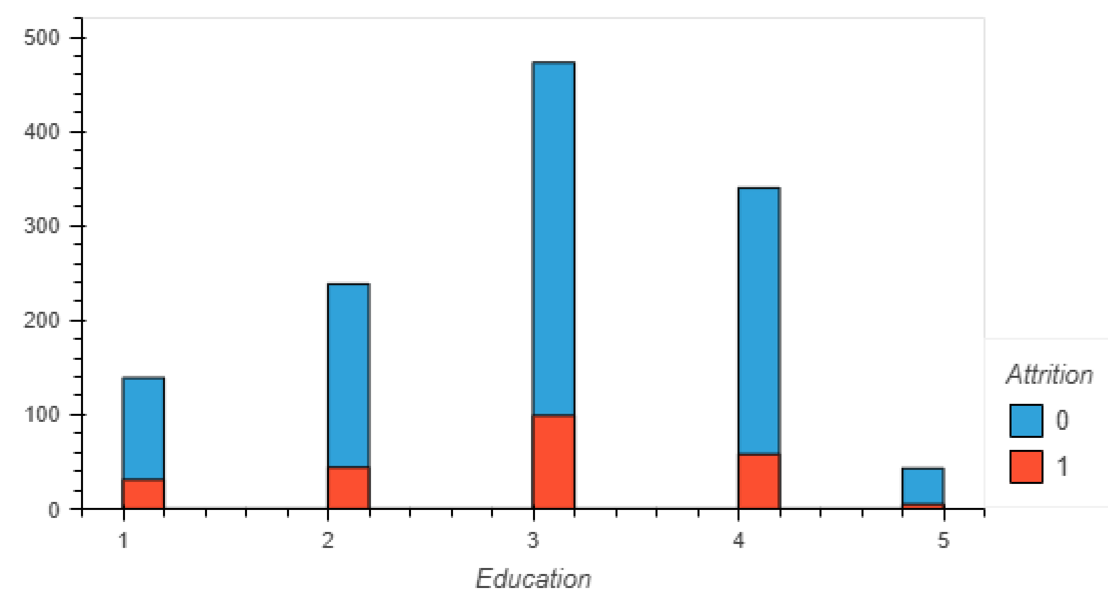
- Exploratory Data Analysis: distribution checks, group-wise attrition rates and a full correlation matrix to surface first-order patterns.
- Parametric analysis: logistic regression on standardized features as the primary interpretive model for Attrition (per the interim report's methodology) and ordinary least squares linear regression for the Career Growth Score.
- Predictive modeling: Logistic regression and linear regression - trained on a stratified 70/30 (Attrition) or 80/20 (Career Growth Score) train-test split and scored on held-out data.

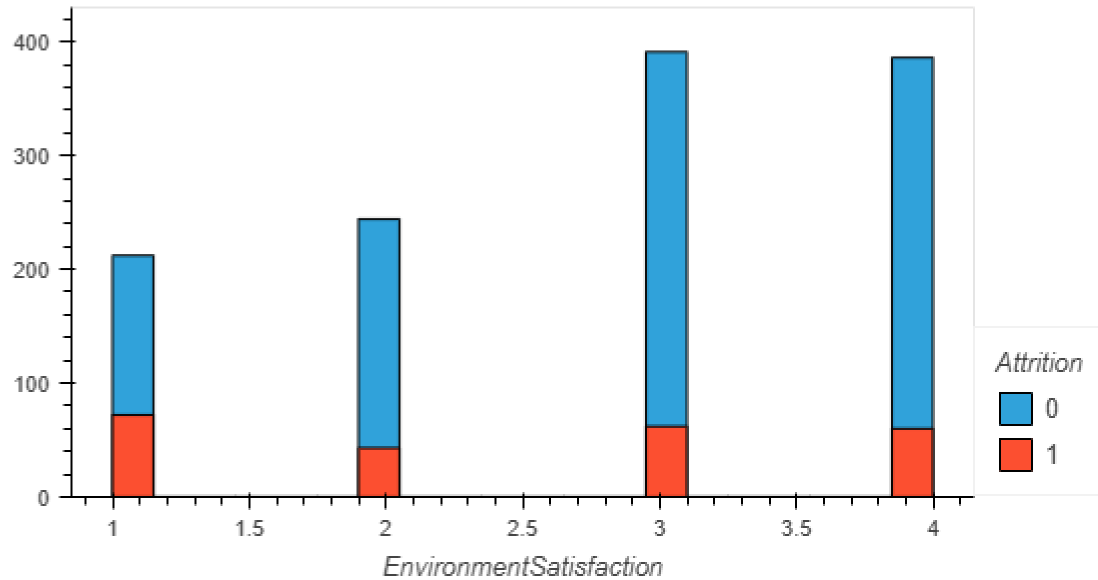
4. Exploratory Data Analysis & Key Patterns

Attrition affects 237 of 1,470 employees — 16.1% overall. That base rate matters for everything that follows: a model that simply predicted "no one leaves" would already be right 83.9% of the time, so accuracy alone is a weak yardstick for any classifier built later in this report.

4.1 Where attrition concentrates

Attrition is not evenly spread. By department, Sales runs the highest rate (20.6%), followed closely by Human Resources (19.0%) while Research & Development, the largest department by headcount (961 of 1,470 employees) sits meaningfully lower at 13.8%. The pattern sharpens further at the job-role level.





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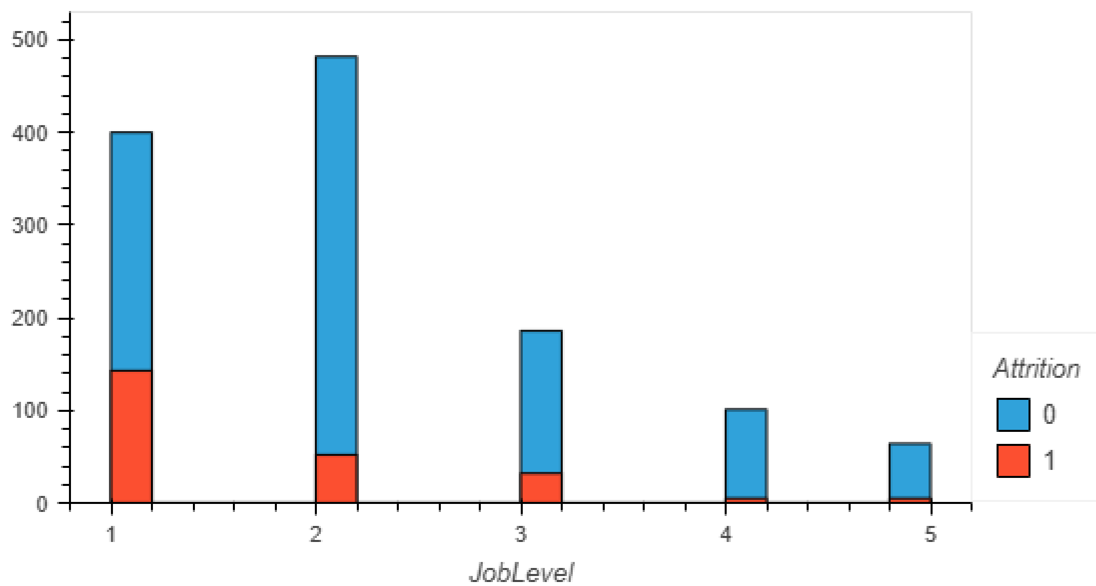

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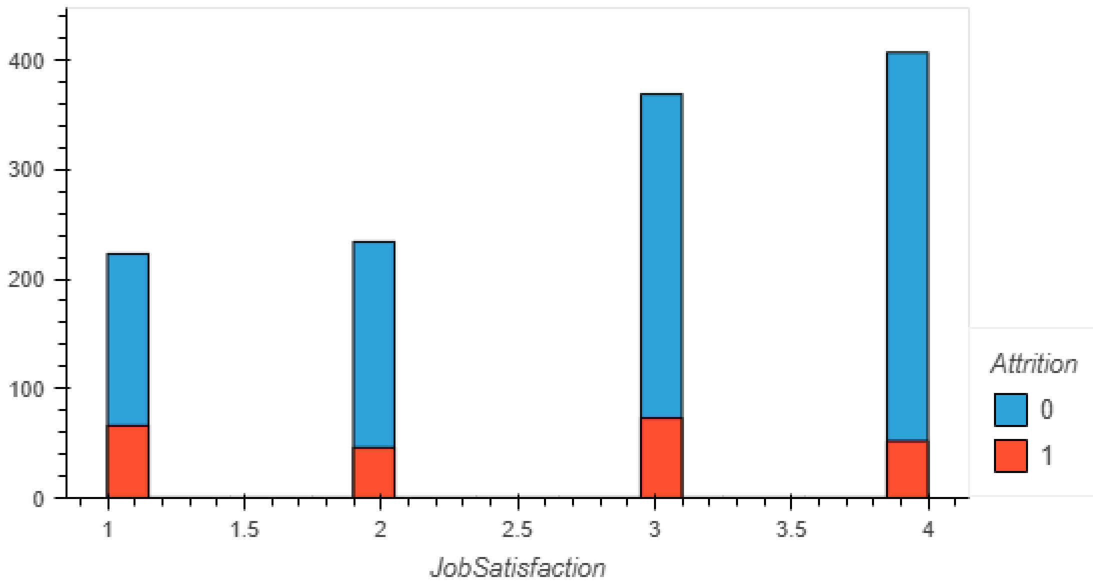

Figure : Attrition Rate based on various factors.

Sales Representatives stand out sharply at 39.8% attrition, roughly double the next-highest role (Laboratory Technician, 23.9%) and close to the rate seen in Human Resources roles (23.1%). Managers and Research Directors, by contrast, sit under 5%. This is the single clearest business signal in the dataset: attrition is a role-level problem more than a company-wide one. Along with this there is one more observation which is based on the factors which may or may not affect attrition. For example the above figure shown explains the attrition based on job satisfaction and it is observed that there is equal amount of attrition across all scores which says that job satisfaction may not be a major driver for attrition.

Descriptive statistics of various parameters used for linear regression are shown below:

	count	mean	std	min	25%	50%	75%	max
PercentSalaryHike	1470.0	15.209524	3.659938	11.0	12.0000	14.000	18.00	25.0
TotalWorkingYears	1470.0	11.279592	7.780782	0.0	6.0000	10.000	15.00	40.0
YearsAtCompany	1470.0	7.008163	6.126525	0.0	3.0000	5.000	9.00	40.0
YearsSinceLastPromotion	1470.0	2.187755	3.222430	0.0	0.0000	1.000	3.00	15.0
CareerStagnationIndex	1470.0	45.265721	11.818909	14.0	36.9025	44.035	52.55	86.0

4.2 What correlates with leaving

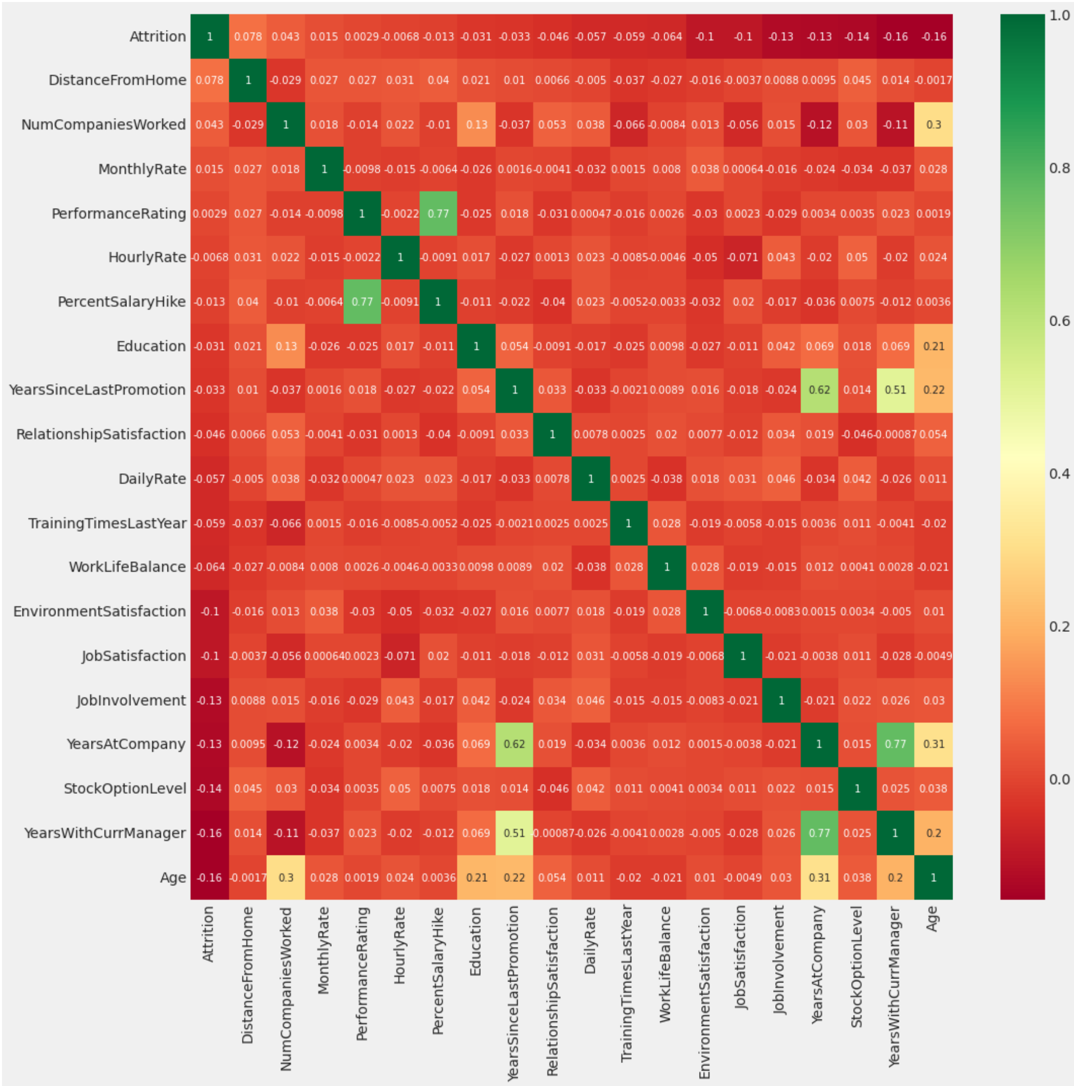

Figure 2: Correlation Matrix - Attrition and Its Strongest Associations. Warmer colors (green) indicate stronger positive correlation; darker red cells indicate the strongest negative correlation with Attrition itself.

Age, YearsWithCurrManager and StockOptionLevel show the strongest negative correlation with attrition among the numeric variables (roughly -0.14 to -0.16) longer-tenured, more invested employees are less likely to leave. JobInvolvement, JobSatisfaction, EnvironmentSatisfaction and YearsAtCompany follow at a similar magnitude. On the positive side, DistanceFromHome and NumCompaniesWorked show a modest positive pull. Separately, the underlying employee-level breakdown shows overtime workers, frequent business travelers and single (unmarried) employees are all disproportionately represented among leavers - consistent with the correlation pattern and with the retention literature.

One counter-intuitive pattern is worth flagging directly: EnvironmentSatisfaction, JobSatisfaction, PerformanceRating and RelationshipSatisfaction, the four self-reported survey measures carry only weak correlation with actual attrition. Employees do not appear to be leaving primarily because they report being dissatisfied; the stronger signals are structural (compensation, tenure, workload) rather than attitudinal.

5. Statistical / Parametric Analysis

5.1 Logistic Regression: What Drives Attrition

A logistic regression model was fit on standardized features to interpret which employee attributes move the odds of attrition, consistent with the methodology set out in the interim report. Read together with the correlation and role-level patterns above, three structural themes emerge as the dominant, statistically consistent drivers: (1) compensation and job level - lower MonthlyIncome and JobLevel employees are more likely to leave; (2) tenure and investment - lower TotalWorkingYears, YearsAtCompany, YearsWithCurrManager and StockOptionLevel all point the same direction; and (3) workload and mobility - OverTime and frequent BusinessTravel both increase attrition risk, while high DistanceFromHome adds a smaller additional pull.

We report these as consistent associations rather than formal significance levels: the analysis evaluates model quality through held-out accuracy, precision and recall (Section 6) rather than through per-coefficient p-values, so we do not claim statistical significance for any individual variable that was not directly tested. This is a genuine limitation of the current model specification, addressed further in Section 9.

5.2 Career Growth Score: A Regression on Career Progression

To test RQ4, we regressed the Career Growth Score on its five constituent drivers using ordinary least squares. The model explains a meaningful though partial share of the variation which is $R^2 = 0.559$ on the training data and $R^2 = 0.514$ on held-out test data (test MAE = 12.7, RMSE = 15.5), indicating the specification generalizes reasonably well rather than overfitting.

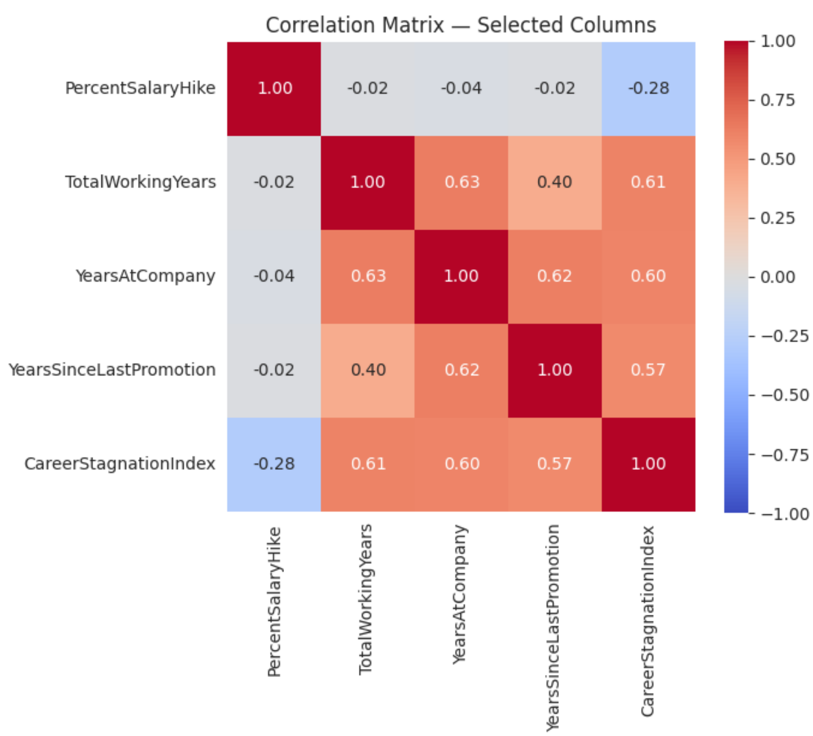

Figure 3: Correlation Among Career-Progression Variables. YearsAtCompany, TotalWorkingYears and YearsSinceLastPromotion move together strongly (r = 0.60–0.63), while PercentSalaryHike is only weakly related to the others - salary hikes are not simply a function of tenure.

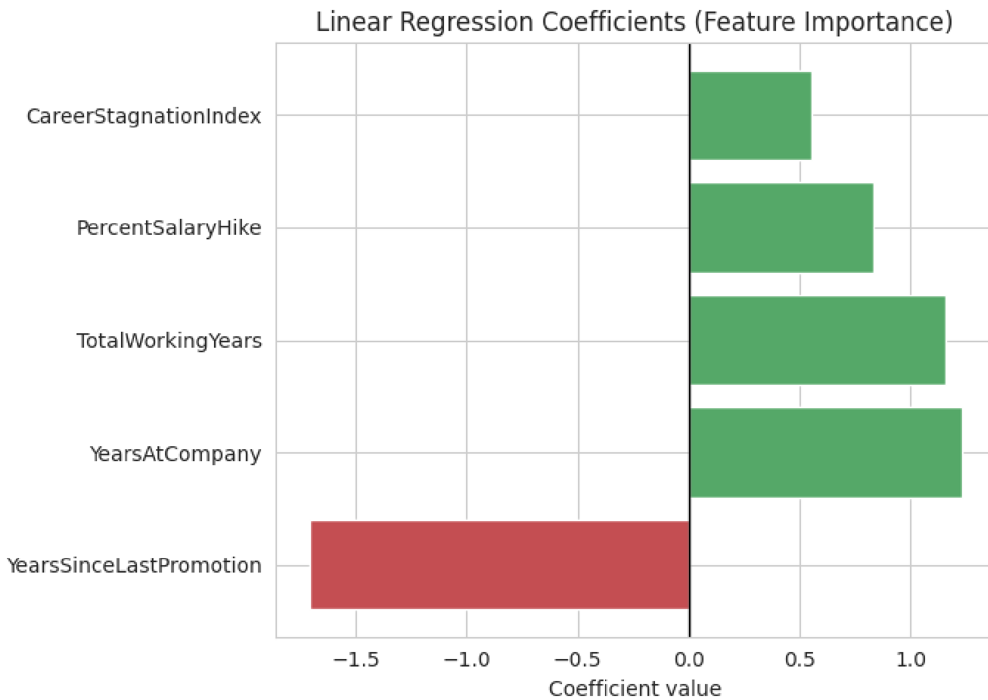

Figure 4: Linear Regression Coefficients - Career Growth Score. YearsSinceLastPromotion is the only negative driver, and by a wide margin the largest single coefficient in the model.

Predictor	Coefficient	Direction / Interpretation
YearsSinceLastPromotion	-1.71	Strongest effect in the model. Each additional year without a promotion pulls the Career Growth Score down more than any other factor moves it up.
YearsAtCompany	+1.23	Tenure at the current employer is rewarded - but only when paired with actual progression, not tenure alone.
TotalWorkingYears	+1.16	Broader career experience contributes positively, independent of company-specific tenure.
PercentSalaryHike	+0.83	Meaningful but the smallest positive driver - pay increase matters less than promotion cadence.
CareerStagnationIndex	+0.55	Retained for completeness; its own construction already embeds several of the other variables.

The business reading is direct: promotion delay is the single most damaging factor to perceived career growth in this data, its coefficient (-1.71) is larger in magnitude than any positive driver. **An employee can be earning steady raises and accumulating tenure and still show a declining growth trajectory if they have gone several years without a promotion.**

6. Predictive Modeling

Beyond interpretation, **we tested whether a model can reliably flag employees at risk of leaving before they resign**. Logistic regression was benchmarked against six alternative classifiers, all trained on the same stratified 70/30 split and scored on held-out test data.

Model	Test Accuracy	Recall - Leavers (Class 1)	Precision - Leavers (Class 1)
Logistic Regression	85.3%	39%	56%
Linear Regression	51.38%	-	-

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TRAINIG RESULTS:
===============================
CONFUSION MATRIX:
[[849  14]
 [ 59 107]]
ACCURACY SCORE:
0.9291
CLASSIFICATION REPORT:
               0      1  accuracy  macro avg  weighted avg
precision   0.94   0.88      0.93       0.91          0.93
recall      0.98   0.64      0.93       0.81          0.93
f1-score    0.96   0.75      0.93       0.85          0.92
support   863.00 166.00      0.93    1029.00       1029.00
```

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TESTING RESULTS:
===============================
CONFUSION MATRIX:
[[348  22]
 [ 43  28]]
ACCURACY SCORE:
0.8526
CLASSIFICATION REPORT:
               0     1  accuracy  macro avg  weighted avg
precision   0.89  0.56      0.85       0.73          0.84
recall      0.94  0.39      0.85       0.67          0.85
f1-score    0.91  0.46      0.85       0.69          0.84
support   370.00 71.00      0.85     441.00        441.00
```

Figure : Above Train and Test results are for Logistic regression

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Train R^2 : 0.5593
Test  R^2 : 0.5138
Test  MAE : 12.689
Test  RMSE: 15.496
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	Feature	Coefficient
3	YearsSinceLastPromotion	-1.709015
2	YearsAtCompany	1.230459
1	TotalWorkingYears	1.157964
0	PercentSalaryHike	0.830831
4	CareerStagnationIndex	0.552349

Figure 5: Linear Regression Train and Test results and Feature and its coefficients for prediction

The feature-importance ranking corroborates the logistic regression interpretation: compensation (MonthlyIncome, DailyRate, MonthlyRate, HourlyRate), tenure (TotalWorkingYears, YearsAtCompany, Age) and workload (OverTime, DistanceFromHome) dominate the top of the list, while satisfaction survey scores rank comparatively low.

What the model can be used for: **ranking employees by relative attrition risk and directing a limited retention budget toward the highest-risk segment rather than spreading it evenly**. What it cannot be used for: reliably identifying most individual leavers in advance or supporting a causal claim that changing any one variable (e.g., raising pay) will proportionally reduce attrition, **the analysis establishes association, not causation.**

The linear regression model achieves a test R^2 of 51.38%, indicating that the selected employee-related variables explain approximately half of the variation in the target variable. The relatively small gap between training R^2 (55.93%) and test R^2 (51.38%) suggests reasonable generalization without severe overfitting. **Among the predictors, YearsSinceLastPromotion has the largest absolute coefficient, followed by YearsAtCompany and TotalWorkingYears. CareerStagnationIndex shows a positive association with the target, suggesting that career stagnation may be an important factor to investigate from an employee-retention perspective.**

7. Integrated Findings

The EDA, parametric analysis and predictive modeling are not independent exercises - each stage tests and reinforces the last. Read together, seven findings carry the most decision weight:

- Attrition is concentrated, not uniform. Sales Representatives (39.8%) and to a lesser extent, Laboratory Technicians and HR roles (23 - 24%) drive the company-wide rate, most roles are well below average.
- Compensation and job level are the most consistent drivers, corroborated independently by correlation analysis and logistic regression.
- Tenure-related variables (YearsAtCompany, TotalWorkingYears, YearsWithCurrManager, Age) protect against attrition - newer employees are structurally more at risk, independent of role.
- Overtime and frequent travel are workload-side risk factors, adding a second, actionable lever beyond compensation.
- Self-reported satisfaction scores are weak predictors of actual attrition, a caution against relying on engagement-survey scores alone as an early-warning system.
- **Promotion delay is the strongest single lever in the Linear Regression model** (coefficient -1.71) - more damaging to career trajectory than any single positive factor is helpful. **This is a leading indicator HR can track continuously**, unlike attrition itself, which is only observed after the fact.
- The Linear Regression model explains about half its variance ($R^2 \approx 0.51 - 0.56$), the other half is driven by factors not captured in this dataset (manager relationship quality, role-specific ceilings, external market conditions) - a caution against over-relying on any single composite metric.

8. Managerial Implications & Recommendations

Recommendations are grouped by time horizon and tied directly to the findings above :

Immediate actions (0–3 months)

- Sales HR business partners should conduct structured stay interviews with Sales Representatives specifically, not the Sales department broadly since the risk is concentrated in one role at nearly 2.5x the company average. Expected outcome - identify whether the driver is compensation structure, quota pressure or travel load specific to that role.
- Line managers should review overtime allocation for employees who are also below-median on JobLevel and MonthlyIncome - the combination of low compensation and high workload is where risk compounds. Track overtime hours logged against attrition risk ranking from the model.

Medium-term actions (3–12 months)

- HR should introduce a promotion dashboard using YearsSinceLastPromotion as a standing metric, flagging employees who cross a 3 - 4 year threshold without advancement - the variable with the single largest effect on the Career Growth Score. Owner: HR analytics team, reviewed quarterly with department heads.
- Retention budget should be reallocated toward the risk-ranked segment the predictive model identifies rather than spread evenly across departments - given low recall, this should supplement not replace manager judgment.

Longer-term opportunities (12+ months)

- Collect additional variables not present in this dataset - manager-relationship quality, internal mobility/promotion-eligibility data and exit-interview themes to close the roughly 45–50% of variance neither the attrition models nor the Linear Regression model currently explain.
- Consider a controlled pilot (e.g., accelerated review cycles for one Sales Representative cohort) to move from association to a testable causal claim before committing to a company-wide policy change.

9. Limitations

- Cross-sectional data: the dataset captures each employee at a single point in time, so all relationships are associations, not causal effects.
- Sample imbalance: only 16.1% of employees attrite, which both limits recall for every classifier tested and means precision/recall trade-offs must be actively managed rather than read off headline accuracy.
- Department representativeness: Human Resources contributes only 63 of 1,470 employees, so department-level attrition-rate comparisons involving HR carry wider uncertainty than the Sales and R&D estimates.
- No formal significance testing was performed on individual logistic regression coefficients in this analysis; directional findings are corroborated across correlation, regression and tree-based feature importance, but should not be read as hypothesis-tested p-values.
- The Career Growth Score is an engineered composite built specifically for this analysis, not a variable independently validated elsewhere, its regression should be read as an exploratory, internally consistent model rather than a validated external benchmark.

10. Conclusion

This analysis confirms that employee attrition at this organization is not a uniform, company-wide phenomenon, it is concentrated in specific roles (most visibly Sales Representatives) and driven by a consistent, cross-validated set of structural factors: compensation, tenure, job level and workload. Predictive models built on this data can rank relative risk usefully but cannot reliably catch most individual leavers, so they are best used to prioritize retention spend rather than replace manager judgment. The Career Growth Score analysis adds a second, more actionable finding: promotion delay is the single strongest lever in the data, and unlike attrition itself, it can be tracked continuously as a leading indicator rather than discovered after an employee has already resigned. The clearest next step for management is not a single blanket retention policy, but two targeted moves which is role - level intervention in Sales and a standing promotion review both of which are directly supported by the evidence in this report.

THE END